Dossier Space Pilot — Design Rationale Note (feeds ADR-004)

Direct Access vs Curation: Why the Custody Server Reads the Raw Source

3 August 2026 · Ken Tombs (principal, whose insight drives this note) with Claude (design lead, Opus 4.8) · Captured during the ADR-004 design discussion, Session 7, to protect the reasoning before it is compressed into the ADR's decision statement. → reference (Master Reference: Architecture).

1. The question that prompted this

Designing the Phase 2 custody server, a fork appeared: should the server's tools reach the corpus DIRECTLY, or only serve files from a pre-approved MANIFEST (a curated list of "the right files, in the right state")? The manifest wall offers a strong-looking scoping guarantee — not in the manifest, does not exist to the AI — and was, on first framing, the more cautious option.

2. The principal's insight

The insight that redirected the design, in the principal's own reasoning: if the corpus is the source / raw data, then by far the best quality is to work DIRECTLY with it — nothing curated sits between the AI and the evidence as it actually is. But if the corpus has been manipulated for some reason, we must FLAG that. And critically: introducing a "sense check" — that if files APPEAR correct then they are correct — is problematic.

The force of this is that a manifest of "approved" files IS exactly such a sense check, institutionalised. Someone decided these files are the right ones, in the right state, and the AI then works within that judgement as though it were ground truth. If the corpus was manipulated BEFORE the manifest was built, the manifest encodes the manipulation AS APPROVED. The wall does not protect the evidence; it launders whatever state the evidence was in when the wall went up. An assertion you cannot interrogate is worth less than a raw source you can.

This is the pilot's navigable-chain doctrine applied to architecture: the manifest ASSERTS a custody state; direct access lets you INTERROGATE it. Curation is precisely where distortion hides, because curation is a human judgement made once and unverifiable afterward.

3. The refinement: direct, but never unrecorded

Direct access to the source is right — with one guard rail, or its quality becomes its weakness. "Direct" must not mean "unrecorded." The danger is not the directness; it is a server that reads the raw corpus without hashing and logging what it read — the highest-quality access married to the weakest custody trail. The posture that honours the insight is therefore: direct read of the raw source, with every read hashed and logged AT THE MOMENT OF ACCESS, so the record captures the evidence exactly as it was when the AI touched it.

Under this posture, manipulation becomes visible rather than assumed away. Manipulation BEFORE the moment of access shows up as a discrepancy against any prior state we independently hold (a pre-existing manifest, a recovered hash). Manipulation AFTER shows up in verify_hash. The server never decides what is authentic; it records what it saw, precisely, so authenticity can be argued from evidence later.

4. The three postures, stated for the ADR

The manifest wall — the server serves only pre-approved files. REJECTED: it curates, and curation hides distortion; it institutionalises a sense-check; it launders the corpus's state-at-approval as authentic.

The naive direct — the server reads raw files and returns them, no hashing. REJECTED: highest-quality access, weakest custody; directness without record.

The recorded direct — the server reads the raw source directly, hashes and logs every access at the moment of access, and provides verify_hash against independently-held prior hashes. ADOPTED: the quality of the raw source, plus a custody trail that can SURFACE manipulation rather than assume its absence.

5. The architectural feature the insight earns: flag, don't curate

The principal's manipulation point earns its own feature, not merely the rejection of the manifest. On first access to the corpus, the server compares the current state against any prior recorded state it has been given — and FLAGS divergence. Not to block it; to surface it. The distinction is the whole philosophy: a wall would have silently refused; a flag reports and lets the humans reason about it.

The Enron corpus is the live proof of concept. It WAS manipulated — the 3 July PowerShell rename that appended .eml to ~517,000 files. We DO hold the pre-manipulation record — manifest.txt (1 July, pre-rename filenames) and the recovered git commit declaring the CMU source. The honest architecture flags "current filenames diverge from manifest.txt — a transformation has occurred here" rather than silently working with the renamed state as though it were original. A server that flags its own corpus's rename is doing exactly what the insight demands: it does not sense-check the evidence into looking correct; it reports what it can and cannot vouch for.

6. Open design question (for the ADR to settle)

Should provenance-divergence flagging be a TOOL the AI calls (e.g. check_provenance, comparing current state against a held baseline), or a BEHAVIOUR baked into the logging layer that runs automatically at access, independent of whether the AI chooses to check?

The design lead's recommendation, for the principal's decision: the latter — baked in, automatic, not AI-invoked. Provenance checking that depends on the AI choosing to run it reintroduces exactly the discretion the custody layer exists to remove; an integrity guarantee the subject can decline to exercise is not a guarantee. Baking it into the logging layer means every access is measured against known baselines whether or not anyone asked — the infrastructure vouches, not the analyst. This mirrors the pilot's standing lesson that governance must be enforced by structure, not by vigilance or discretion.

7. Why this note exists

This reasoning is compressed, in ADR-004, into a decision statement of a few lines. That compression is correct for a decision record but lossy for the THINKING, and the thinking here is significant: it establishes that in evidence work, direct interrogable access to raw source, fully recorded, is a stronger custody posture than curated access — because curation is an unverifiable assertion and rawness is an interrogable fact. The note preserves the derivation so that a reviewer, an assurer, or a future operator can see not just what was decided but why the obvious-seeming safer option (the wall) was the weaker one. → Master Reference: Architecture / Custody posture.